VICTRON VENUS OS · CAMPER VAN

Tank sensors & fridge for Venus OS

Step-by-step guide: wiring, ESP32 firmware upload, tank calibration and installation on a Raspberry Pi running Venus OS.

Mopeka Check Standard / XL — LPG bottle level

ESP32-S3 hydrostatic sender — fresh-water level for non-square tanks, water temperature

Bluetooth compressor fridge — fridge and freezer temperature (Alpicool controller; check Hyckes first)

github.com/mich-crypto/Venusosmods · generated 2026-10-27

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Before you start: this is a do-it-yourself project, not a Victron product. Work on the 12 V system with the fuse removed, and keep drinking-water parts food-safe. The steps have been tested in simulation. Each part of the guide ends with a check, so confirm it on your own installation before moving on.

Installation guide

This guide takes you from parts on the table to tanks and the fridge on your Venus OS screen and in VRM. Work through it in order. Each part ends with a check, so you know it works before moving on.

Part	What	Time
1	Prepare Venus OS (root access, SSH, Bluetooth)	15 min
2	Build and wire the ESP32 water sensor	1–2 h
3	Upload the firmware to the ESP32	20 min
4	Calibrate the water tank	30 min
5	Install the drivers on Venus OS	20 min
6	Mopeka gas sensor	10 min
7	Fridge	10 min
8	Updates, troubleshooting, removal	–

You need: the Raspberry Pi running Venus OS on the same network (Wi-Fi or LAN) as a laptop, a USB-C **data** cable (some cables only charge), and the parts from wiring.md.

Part 1 – Prepare Venus OS

1. Open the Remote Console: on the Pi's screen, or in a browser at `http://venus.local` (or the Pi's IP address).
2. **Enable superuser access.** In *Settings* → *General*:
    1. Set *Access level* to *User & Installer* (password `ZZZ`).
    2. Highlight *Access level* without opening it, then press and hold the **right arrow** key for 5 seconds. *Superuser* now appears.
    3. Choose *Superuser*, then *Set root password*.
    4. Turn on *SSH on LAN*.

    Menus differ slightly between Venus OS versions. Victron's "Venus OS: Root Access" documentation has the details for each.
3. **Bluetooth:** *Settings* → *Bluetooth* → enabled.
4. **Check:** on the laptop, open a terminal (Windows: PowerShell) and run:

    ```
    ssh root@venus.local
    ```

    If `venus.local` isn't found, use the IP address from *Settings* → *Ethernet / Wi-Fi*. You should get a `root@raspberrypi...:~#` prompt. Type `exit` to leave.

Part 2 – Build and wire the ESP32 water sensor

Follow wiring.md. In short:

1. Set the buck converter to **5.0 V** with a multimeter *before* connecting anything to it.
2. Wire buck → diode → ESP32 `5V`, plus GND.
3. Connect the pressure sensor (option A: voltage, or B: 4–20 mA) to GPIO4 through the resistors.
4. Optional: DS18B20 to GPIO5 with the 4.7 kΩ pull-up to 3V3.
5. Don't put the sensor in the tank yet; it's easier to test on the table.

Check: with the 12 V on, the ESP32's power LED is on and the buck converter still reads 5.0 V (about 4.7 V after the diode).

Part 3 – Upload the firmware to the ESP32

3.1 Install the Arduino IDE (once)

1. Download **Arduino IDE 2** from arduino.cc and install it.
2. *File → Preferences → Additional boards manager URLs*, add:

   ```
   https://espressif.github.io/arduino-esp32/package_esp32_index.json
   ```

3. *Tools → Board → Boards Manager*: search **esp32**, install **esp32 by Espressif Systems** (3.x; 2.0.17 works too).
4. *Tools → Manage Libraries*: install **OneWire** (Paul Stoffregen) and **DallasTemperature** (Miles Burton). Only needed with the DS18B20.

3.2 Open and configure the sketch

1. Download this repository (GitHub: *Code → Download ZIP*) and unzip it.
2. *File → Open →* `firmware/esp32_hydro_tank/esp32_hydro_tank.ino`.
3. In the **CONFIGURATION** block at the top, set:
   - `INPUT_CURRENT_LOOP` and the `SIGNAL_*` values for your sensor (the values for both options are in wiring.md)
   - `SENSOR_RANGE_MM`: the sensor's full scale in mm of water. From the datasheet: 1 m H_2O = 1000 mm, 10 kPa ≈ 1020 mm, 0.1 bar ≈ 1020 mm.
   - `USE_DS18B20`: `1` with the probe, `0` without
   - `DEVICE_ID`: leave at `1` unless you build a second sender
   - `DEFAULT_TABLE`: leave as it is. You calibrate later over USB, without editing code.

3.3 Board settings (*Tools* menu)

Setting	Value
Board	**ESP32S3 Dev Module**
USB CDC On Boot	**Enabled** (otherwise you see no serial output)
Flash Size	match your board (4 MB / 8 MB / 16 MB; 4 MB always works)
Partition Scheme	Default
Upload Speed	921600 (use 115200 if uploads fail)
Port	the new COM / `/dev/ttyACM*` / `/dev/cu.usbmodem*` port

Many S3 boards have **two** USB-C ports, labelled *USB* and *UART* (or *COM*). Either works for uploading. With *USB CDC On Boot: Enabled* the Serial Monitor works on the **USB** port. On the *UART* port, set *USB CDC On Boot* to *Disabled*.

3.4 Upload

1. Unplug the buck converter's 5 V if you didn't fit the diode.
2. Connect the ESP32 by USB, choose the port, click **Upload** (→).
3. If it hangs at `Connecting....`: hold the **BOOT** button, tap **RST**, release BOOT, and upload again. After uploading, tap **RST** once.

3.5 Check

Tools → *Serial Monitor*, **115200 baud**, line ending **Both NL & CR**. You should see:

```
Using the default tank table from the sketch (no calibration stored)
Advertising as VHT, MAC 34:85:18:xx:xx:xx
Type 'help' for calibration commands.
signal=0.512 height=3.0mm litres=0.4 level=0.5% temp=19.6 activity=0.4
```

- `signal` should be close to `SIGNAL_AT_ZERO` with the sensor in air (0.5 V or 4 mA).
- Dip the sensor into a bucket: `height` should rise by roughly the depth.
- `FAULT` = signal outside the fault limits: check wiring and the `SIGNAL_*` settings.
- `temp=nan` = no DS18B20 found (wiring, pull-up, or `USE_DS18B20 0`).

Note the **MAC** for later (optional in config.ini).

Part 4 – Calibrate the water tank

Install the sensor in the tank, then follow calibration.md. Short version, with the van level and the tank empty:

```
zero
cal start
   (pour 10 L in, wait 30 s)
cal add 10
   (repeat until full)
cal save
```

No measuring container? Use the dimension calculator in calibration.md, method B.

Part 5 – Install the drivers on Venus OS

5.1 Copy the files

From the folder where you unzipped the repository:

macOS / Linux / Windows PowerShell:

```
scp -r data/shared_drivers data/etc root@venus.local:/data/
```

On Windows you can also use **WinSCP** (SCP protocol, user `root`): copy the folders `data/shared_drivers` and `data/etc` into `/data/` on the Pi.

5.2 Create the configuration

```
ssh root@venus.local
sh /data/shared_drivers/install.sh
```

The first run creates `/data/shared_drivers/config.ini` and stops.

5.3 Find your sensors

```
python3 /data/shared_drivers/diagnostics/test_venus_compatibility.py --scan 30
```

Press the Mopeka's sync button during the scan. The scan lists:

- `PASS Mopeka Standard AA:BB:...` → put that MAC in `[mopeka_standard]`
- `PASS ESP32 tank id 1 ...` → the water sensor is received (MAC optional)
- `PASS Fridge candidate ...` → see Part 7

The upper part of the output checks the environment (Python, D-Bus, velib_python, Bluetooth, config). Fix any **FAIL** before continuing.

5.4 Edit config.ini

```
nano /data/shared_drivers/config.ini
```

(`Ctrl+O`, `Enter` saves, `Ctrl+X` quits.) Set the MACs, the bottle size and `propane_ratio` for the Mopeka, and names. Every option is explained in the file. Delete or ignore sections for sensors you don't have.

5.5 Enable the services

Only the ones you use:

```
sh /data/shared_drivers/install.sh mopeka_standard esp32_hydro fridge
```

This runs the diagnostics first and refuses to enable anything if one reports FAIL. It links the services into `/service` and adds them to `/data/rc.local` so they come back after reboots and Venus OS updates.

5.6 Check

- Remote Console: the **tanks** appear on the overview page and under *Device list*. The water temperature and fridge zones appear as temperature sensors in the *Device list*.
- Logs:

  ```
  tail -F /var/log/esp32_hydro/current | tai64nlocal
  ```

- Live D-Bus values: `dbus-spy` (arrow keys, `Enter` opens a service, `q` quits).
- **VRM:** after a few minutes, the tanks and temperatures appear on the VRM dashboard and in *Advanced* widgets. To rename a device, use its *Device* page on the GX; VRM picks the new name up.
- Reboot the Pi once (`reboot`) and check everything comes back.

Part 6 – Mopeka gas sensor

1. Put the sensor on the **centre** of the bottle's bottom, with plenty of coupling gel (or the supplied pad).
2. In `dbus-spy` , open `com.victronenergy.tank.mopeka_standard` and watch `/Mopeka/DistanceMm` and `/Mopeka/EchoQuality` .
   - EchoQuality stays low (<3) or the distance jumps around: re-seat the sensor, use more gel, clean the bottle bottom.
3. Note the distance with a nearly **empty** bottle → `empty_mm` , and with a freshly **filled** bottle → `full_mm` . Gas bottles are filled to about 80 %, so "full" is what a new bottle reads.
4. `propane_ratio` : 1.0 for pure propane (most of Northern Europe, and winter gas), around 0.3–0.4 for summer propane/butane mixes (e.g. southern Europe). This corrects the speed of sound in the liquid.
5. `capacity_l` : liquid volume, e.g. 11 kg ≈ 27 L, 5 kg ≈ 12 L.
6. `svc -t /service/mopeka_standard` restarts the driver after changes.

Part 7 – Fridge (Hyckes and other Bluetooth compressor fridges)

Many 12 V compressor fridges use the same Bluetooth controller, the one the "Alpicool / CAR FRIDGE FREEZER" app talks to. The driver reads the temperatures of both zones, the set points, the supply voltage and the eco/power state. It does **not** change any fridge settings.

Hyckes's own app could not be checked here, so first test whether your fridge uses this controller:

1. **Close the fridge app on your phone**, or switch off the phone's Bluetooth. The fridge accepts only one connection at a time.
2. On the Pi:

```
python3 /data/shared_drivers/diagnostics/fridge_probe.py
```

Look for `→ LIKELY FRIDGE` (names like `WT-0001`, `A1-...`, `AK1-...`). No candidate? Note the devices with the strongest signal while standing next to the fridge.

3. Test it:

```
python3 /data/shared_drivers/diagnostics/fridge_probe.py AA:BB:CC:DD:EE:FF
```

   - `OK - this fridge speaks the Alpicool protocol` plus temperatures: put the MAC in `[fridge]` in config.ini and enable it with `install.sh fridge`.
   - `characteristics ... not found`: Hyckes uses a different controller. Keep the printed *GATT characteristics* list: with it (and ideally a Bluetooth capture from the phone app), the protocol module can be extended.

Sharing with the phone app: by default the Pi connects for about 2 s every 60 s (`poll_interval`) and then disconnects, so the phone app can still connect in between. While the phone app is connected, the Pi can't read the fridge; after `timeout` seconds the zones show as disconnected and recover automatically.

Part 8 – Updates, troubleshooting, removal

Updating the drivers

Copy the new files over the old ones (5.1), then restart:

```
svc -t /service/mopeka_standard /service/esp32_hydro /service/fridge
```

`config.ini` is not overwritten (it isn't in the repository).

Venus OS firmware updates

Nothing to do: everything lives in `/data`, and `/data/rc.local` restores the services at boot. Afterwards, run the diagnostics once to confirm.

Troubleshooting

Symptom	Check
Tank not shown	`svstat /service/<name>` should say `up`; read the log in `/var/log/<name>/current`
Service keeps restarting	run it by hand to see the error: `svc -d /service/<n>` then `python3 /data/shared_drivers/<driver>.py --section <n> --debug`
Tank shows "disconnected"	the sensor isn't heard: run `--scan 30`, check the range (metal lockers block Bluetooth)
Water level wrong	redo `zero` and the calibration; check the sensor lies flat and the vent tube is open
Water level slow while driving	intended (anti-slosh), settles once parked
Mopeka level jumps	coupling gel, sensor position, `median_window = 9`
Fridge disconnected	phone app connected? Bluetooth range? `fridge_probe.py <MAC>`
Two devices with the same instance	*Settings → VRM online portal → VRM device instances* (Venus OS 3.x), or `device_instance` in config.ini

Test without hardware: the `tests/` folder contains a simulation that runs all drivers against a fake Bluetooth stack (see the README).

Removing

```
sh /data/shared_drivers/uninstall.sh            # all services
sh /data/shared_drivers/uninstall.sh fridge     # just one
```

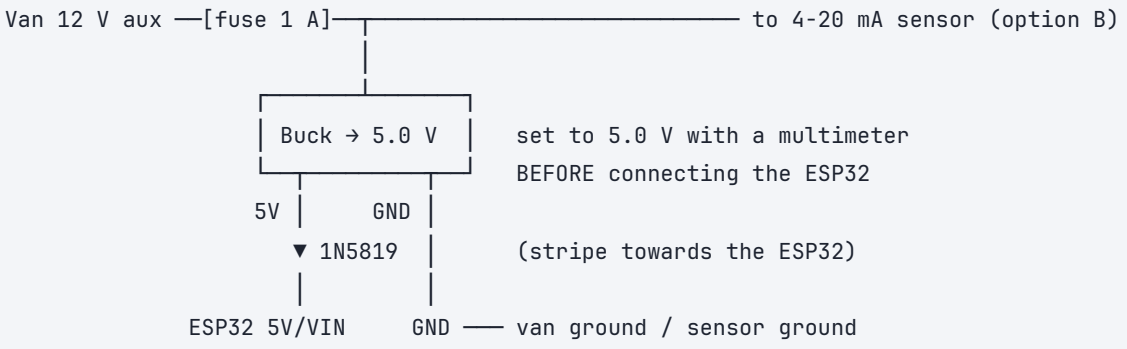
This stops the services and removes them from `/data/rc.local`. Delete `/data/shared_drivers` and `/data/etc/init.d/<name>` to remove the files.

Wiring

Parts

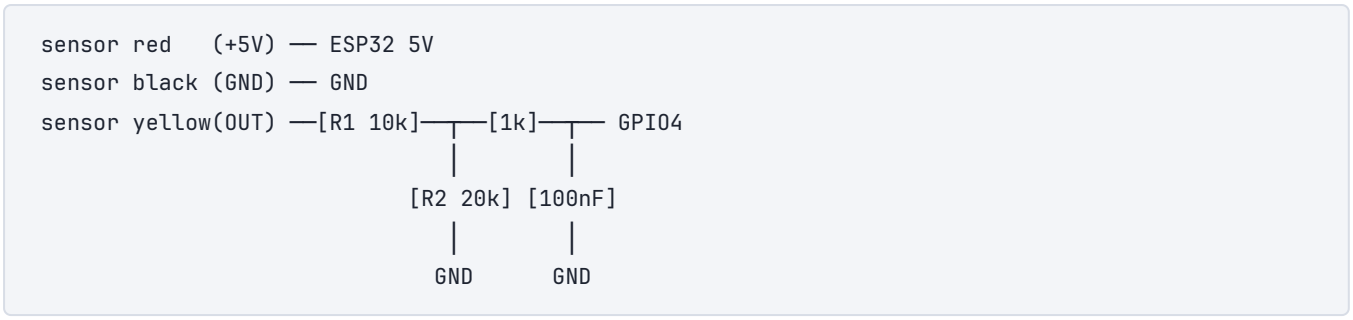
Part	Notes
ESP32-S3 dev board (DevKitC-1 or clone)	any S3 board with USB-C
Buck converter 12 V → 5 V, ≥1 A	MP1584 or LM2596 module, adjustable
Fuse holder + 1 A fuse	at the 12 V tap
Schottky diode 1N5819 / SS14	lets USB and 12 V be connected together
Hydrostatic level transmitter	0.5–4.5 V or 4–20 mA output, range ≈ tank height (see below)
Resistors	voltage version: 10 kΩ + 20 kΩ; 4–20 mA version: 120 Ω 0.1 %/1 %
1 kΩ resistor + 100 nF capacitor	input filter/protection
DS18B20 waterproof probe + 4.7 kΩ	optional water temperature
IP65 junction box, cable glands, ferrules	
Wire	0.75 mm² for 12 V, any thin wire for signals

Power



- Use the board's **5V / VIN** pin, never 3V3.
- The diode drops ~0.3 V (ESP32 sees ~4.7 V, fine) and stops the buck converter and the laptop's USB from feeding into each other while you calibrate. Without it: unplug the 5 V wire before plugging in USB.
- Take 12 V from a circuit that is on when the Victron system is on (e.g. the 12 V distribution after the main switch), not from the starter battery. The ESP32 draws about 0.5 W.

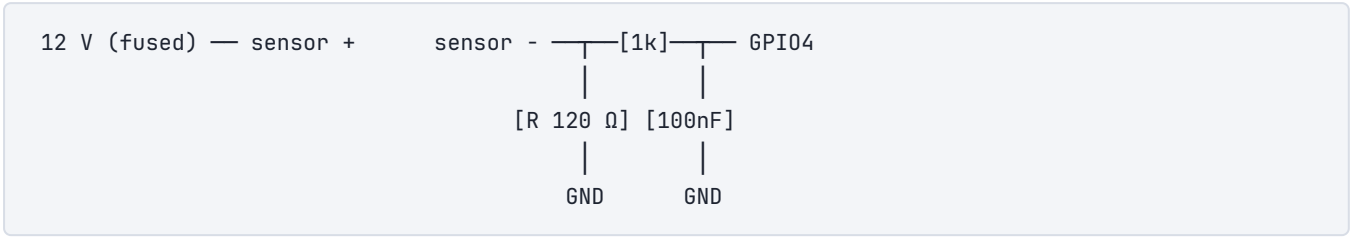
Pressure sensor, option A: voltage output (0.5–4.5 V, 5 V supply)



Divider ratio (R1+R2)/R2 = 1.5, so 4.5 V becomes 3.0 V at the pin. Firmware: `INPUT_CURRENT_LOOP 0`, `DIVIDER_RATIO 1.5f`, `SIGNAL_AT_ZERO 0.5f`, `SIGNAL_AT_FULL 4.5f`, `SIGNAL_FAULT_LOW 0.2f`, `SIGNAL_FAULT_HIGH 4.9f`.

Wire colours differ between makers: check the sensor's label or datasheet.

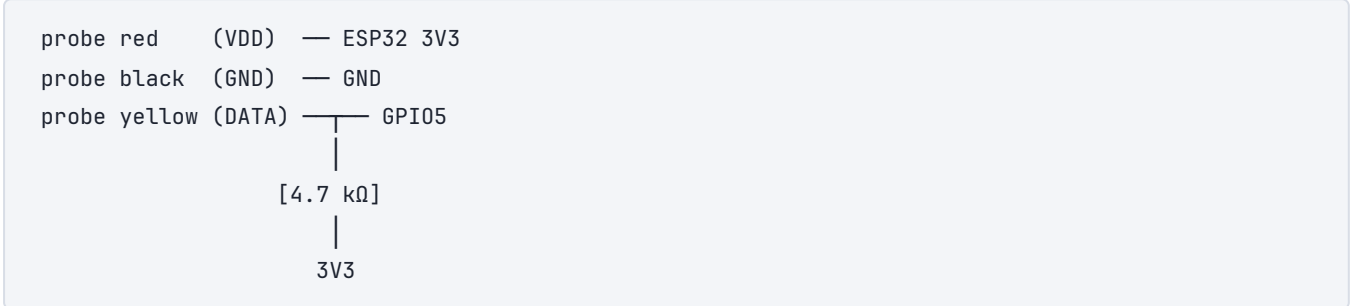
Pressure sensor, option B: 4–20 mA two-wire transmitter (12–24 V supply)



4 mA → 0.48 V, 20 mA → 2.4 V. Use a 0.1 % or 1 % resistor. Firmware: `INPUT_CURRENT_LOOP 1`, `SHUNT_OHMS 120.0f`, `SIGNAL_AT_ZERO 4.0f`, `SIGNAL_AT_FULL 20.0f`, `SIGNAL_FAULT_LOW 3.0f`, `SIGNAL_FAULT_HIGH 22.0f`. A broken wire then shows up as a fault in the GUI instead of "empty". This type copes best with long cables and electrical noise in a vehicle.

The 1 kΩ + 100 nF form an RC filter and limit current into the pin's protection diodes if something goes wrong.

DS18B20 temperature probe (optional)



- Power it from **3V3**, not 5 V: the data line goes straight to the ESP32.
- The 4.7 kΩ pull-up is required. For cables over ~5 m use 2.2 kΩ.
- Set `USE_DS18B20 0` in the sketch if you don't fit one.
- Drinking water: the common "waterproof stainless" probes are not certified for potable water. Either buy one with a documented food-grade sleeve and cable, or put the probe in a dry pocket on the **outside** of the tank wall (clamped with thermal paste and insulated). That measures the water through the wall, a little slower, without touching it.

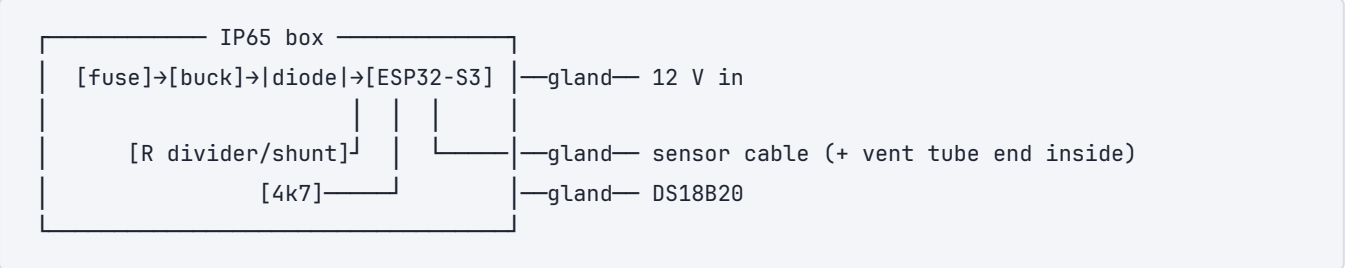
Choosing and mounting the pressure sensor

- **Range:** pick one just above the tank height, e.g. 0–1 m H_2O (0–10 kPa) for a 40 cm tank. A 0–5 m sensor would only use 8 % of its range, so you lose resolution.
- **Vented (gauge) sensor:** the cable has a small vent tube. Keep its end open to air and dry: inside the junction box is fine, outside under the van is not. Don't seal it with the cable gland.
- **Position:** flat on the bottom, as close to the **middle** of the tank as you can. On a slope the level at the centre hardly changes.
- **Drinking water:** stainless (316) body and a food-safe (e.g. FDA/KTW certified) cable, or mount the sensor outside the tank in a T-piece on the drain at the lowest point.
- **Through the lid:** most installs lower the probe through a cable gland (PG9/PG11) in the tank's inspection lid, so no new hole below the water line.

ADC pins

On the ESP32-S3 only GPIO1..GPIO10 are ADC1. ADC2 (GPIO11..20) shares hardware with the radio and can return errors while it is active, so keep the sensor on ADC1. GPIO4 and GPIO5 are the defaults in the sketch (`PIN_PRESSURE`, `PIN_ONEWIRE`).

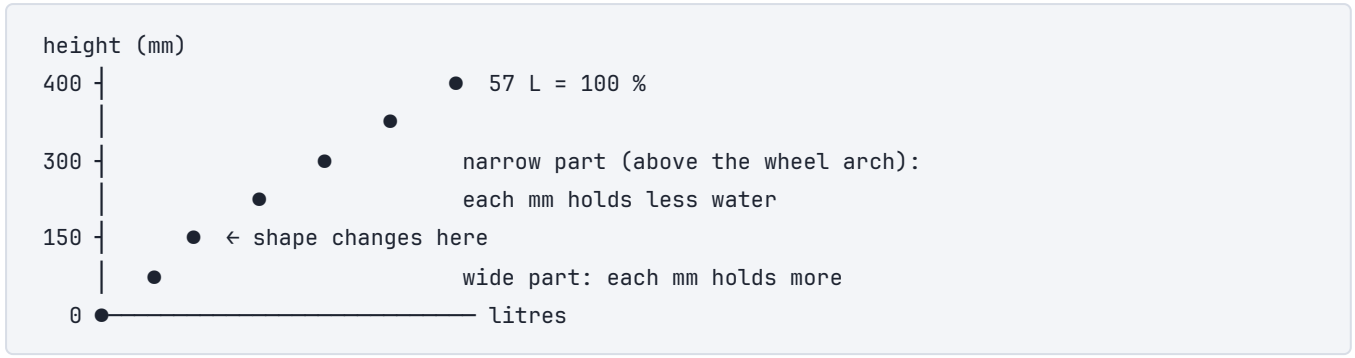
Junction box layout



Mount the box above the tank's maximum water level, with the glands pointing down so condensation drains away from the board. Plastic boxes let Bluetooth through; a metal box or a box inside a metal locker will cut the range.

Calibrating the water tank

A hydrostatic sensor measures **pressure**, which only depends on the **height** of the water above it, not on the tank's shape. In a square box, height and volume are proportional. In a van tank with a wheel-arch cut-out, a slope or rounded corners they are not: the first 5 cm may hold 12 L, the top 5 cm only 4 L. The ESP32 fixes this with a **height → litres table** that it interpolates between.



You do this once. The table is stored in the ESP32's flash and survives power cuts and firmware updates. There are two ways to build it:

- **A. Fill it in steps** (most accurate, ~30 minutes)
- **B. Calculate it from the tank's dimensions** (when you can't easily fill it in measured steps)

Both use the Arduino IDE **Serial Monitor** (115200 baud, "Both NL & CR") with the ESP32 connected to your laptop over USB-C. Only keep the van's 12 V connected at the same time if you fitted the Schottky diode from wiring.md; otherwise unplug the buck converter's 5 V first.

0. Before you start

1. Mount the sensor where it will stay, lying flat on the bottom, as close to the middle of the tank as possible.
2. Park the van **level** (use a spirit level on the floor, or an app).
3. Type `log` to stop the status lines scrolling, or leave them on to watch the height settle.
4. Type `status`: you should see a plausible `signal=` (0.5 V or 4 mA at zero). `FAULT` means a wiring problem; see wiring.md.

1. Zero point (both methods)

With the tank **completely empty** (or emptied down to the drain):

```
zero
```

The ESP32 averages the reading for 5 s and stores an offset so that the height now reads 0 mm. This removes the sensor's offset and the height of its membrane above the tank floor.

2A. Calibrate by filling in steps

You need a container of known size: a 10 L bucket, or a 5 L jerrycan. Use the same one every time, and measure what it really holds once, on a kitchen scale (1 L of water = 1 kg).

```
cal start
```

records 0 L at the current (empty) height. Then repeat:

1. Pour in one container (for example 10 L).
2. Wait **30 seconds** for the water to settle. If `SLOSH` is shown, wait longer.
3. Type the amount you added **in this step**:

   ```
   cal add 10
   ```

   The ESP32 measures for 5 s and stores the point with the running total (10, 20, 30 L...).

Use smaller steps (5 L) where the shape changes, e.g. around the top of the wheel arch. A wrong point can be removed with `cal undo`, and `cal show` prints the table so far.

When the tank is full:

```
cal save
```

The table is now active and stored. The last point is 100 %. `cal save` also prints the table as C code: copy it into `DEFAULT_TABLE` in the sketch as a backup (otherwise `reset`, or an ESP32 full flash erase, loses it).

Tip: fill through the normal filler neck, so the tank's breather and any air pockets behave like they will on the road.

2B. Calculate it from the tank's dimensions

Measure the **inside** of the tank (outside dimensions minus the wall, often 3-5 mm per wall for roto-moulded tanks) at every height where the shape changes:

- width (left–right) and depth (front–back) at that height
- if one of them is the same everywhere, you only need the other one

Example: a 300 mm deep tank that is 600 mm wide for the first 150 mm, then 400 mm wide (wheel arch) up to 400 mm:

```
python3 tools/tank_table.py --depth 300 --profile "0:600 150:600 151:400 400:400"
```

Width and depth both change? Use `height:width:depth`:

```
python3 tools/tank_table.py --profile "0:500:250 100:550:300 380:550:300"
```

Options:

- `--offset 15`: the sensor's membrane sits 15 mm above the floor, for example because of its housing. Water below it can't be measured, so the table starts counting litres from there.
- `--scale 0.97`: correct for internals (baffles, pump, fittings), or to match the volume the tank's datasheet gives.

The script prints `cal point ...` lines. Paste them into the Serial Monitor (after `zero`), followed by `cal save`.

Check: add a known amount (one bucket) and compare. If the reading is consistently off by a few %, re-run with `--scale`.

3. Check and fine-tune

- After calibration, fill and empty a few litres and watch `litres=` in the log. The level should change by what you added, within about 1-2 %.
- Driving: `SLOSH` appears and the reading becomes very slow (~25 s time constant). That is intended; it settles again within seconds when parked.
- Parked on a slope, the reading changes a little if the sensor isn't in the middle of the tank. That is the water level really changing at the sensor; only moving the sensor fixes it.
- Freezing: a hydrostatic sensor can be damaged when the water around it freezes. Drain the tank in frost, or use the van's tank heater.

Command reference

Command	Does
`help`	list commands
`status`	signal, raw and filtered height, litres, active table
`log`	turn the periodic status line on/off
`zero`	empty tank: current height becomes 0 mm (stored)
`zero <mm>`	set the offset by hand
`cal start`	new table, 0 L at the current height
`cal add <L>`	record a point after adding `<L>` litres
`cal point <mm> <L>`	add a point by hand (total litres)
`cal undo` / `cal show` / `cal cancel`	edit the table being measured
`cal save`	activate + store the table, print it as code
`table`	print the active table as code
`reset`	erase stored calibration, use `DEFAULT_TABLE` from the sketch

The Venus OS side needs no changes after calibration: the ESP32 sends level %, litres and capacity, and the tank's capacity on the GX updates automatically.